Settings

The application stores user preferences in `QSettings` under organisation `madrigal`, application `protocol-lab`, grouped in four sections: `general`, `editor`, `paths` and `advanced`. The window reads them on start and applies them live; nothing is written until the user accepts the dialog.

Open the dialog with **View → Settings...** or the **Ctrl+,** shortcut.

General

Setting	Values	Default
Theme	Dark, Light, System	System
Language	Russian, English, System (shown in the interface language)	System

The theme and the interface language apply as soon as they are changed, through the same `ThemeManager` and `LanguageManager` the menus use. `System` follows the desktop colour scheme and locale; see `docs/STYLE.md` for the palettes and the catalogues in `src/ui/locale/`. Both languages ship a compiled catalogue, so the picker and every other caption follow the active language rather than mixing two.

Editor

Setting	Range	Default
Hex font size	6–48 px	13
Tree font size	6–48 px	11
Show offset column	on / off	on

The hex font size scales the byte grid and its column header together. Hiding the offset column removes the address gutter; the hex and ASCII columns keep their alignment and the byte-to-packet lookup keeps working.

The same sizes are reachable without the dialog: **Ctrl +** enlarges and **Ctrl −** shrinks the bytes and the tree together, and **Ctrl 0** restores the defaults above. The keyboard zoom steps the byte size within 8–28 px and keeps the tree two points smaller, so repeated zooming cannot drift the two apart; the dialog range stays wider for a one-off fine adjustment. Both write the same `editor/hex_font_size` and `editor/tree_font_size` keys.

Paths

Setting	Meaning	Default
Projects directory	where save dialogs start	empty
Captures directory	where open dialogs start	empty
Relative paths in manifest	store portable paths	on

A configured directory is used the first time a file dialog opens. Once a file has been opened or saved, its directory is remembered for the rest of the run and takes precedence.

Advanced

Setting	Values	Default
Logging level	DEBUG, INFO, WARNING, ERROR	INFO
Show diagnostics in UI	on / off	on
Show the byte-state legend	on / off	on

The logging level reconfigures the root logger. Turning diagnostics off hides the gap and ambiguity shading in the hex view so the raw bytes read without it; the diagnostics themselves remain attached to the stream and stay reachable through the provenance line. Turning the byte-state legend off hides the strip above the hex view; the bytes keep their colour, only the explanatory row is removed.

Buttons

- **OK** applies the values and closes.
- **Apply** applies without closing, so several sections can be tuned in a row.
- **Cancel** restores whatever was active when the dialog opened and closes without applying.

Storage keys

Group	Key	Type
general	`theme`	string
general	`language`	string
editor	`hex_font_size`	int
editor	`tree_font_size`	int
editor	`show_offset`	bool
paths	`project_dir`	string
paths	`capture_dir`	string
paths	`relative_paths`	bool
advanced	`log_level`	string
advanced	`show_diagnostics`	bool
advanced	`show_legend`	bool

The defaults are declared once in `src/ui/settings.py`; a reset simply clears the `madrigal/protocol-lab` settings scope.